

ActInf Livestream #029.0 ~ "Active Inferants: An Active Inference Framework"

Livestream | 1:36:38

[Music]

hello everybody welcome to the active
inference lab this is live stream number
29.0 on

september 14th is it september 14th not
anymore

september 17

17.

at least where we're at

on september 17 2021

welcome to the active inference lab

we are a participatory online lab that
is communicating learning and practicing
applied active inference you can find us

at the links here on this slide

this is recorded in an archived live

stream so please provide us with

feedback so that we can improve our work

all backgrounds and perspectives are

welcome and we'll be following good

video etiquette for live streams

at the short link on this slide you can

see the past present and future live
streams

and just to call your attention to the

tabs on the bottom

the first tab are our regular tuesday

live streams so for these ones we do the

zero contextualizing video that's what

you're watching now and then we have a

dot one and a dot two where on two

successive weeks we try to have the

authors try to have a lot of different

people on a panel just talking about the

paper

the guest stream is sort of a wild card
slash anything goes we have a lot of
different topics
model streams reflect machine learning
different kinds of modeling and
walkthroughs of code and the math stream
is for talking about math and different
formal frameworks so if you are
interested in co-organizing or
presenting for any of these different
kinds of streams then just get in touch
with us
today in active stream 29
we're going to be having the goal of
learning and discussing this paper
active inference and active inference
framework for ant colony behavior
by freedman chance ramstead fristen and
constant
and just like
all of the other.0 videos this is just
an introduction to some of the ideas
it's not a review
or a final word
and it doesn't change too much even when
the first or any author is on the stream
we're gonna go over the keywords
as well as the aims and claims the
abstract all the usual things that we
cover in the dot zeros and in the coming
weeks we're gonna be discussing this
paper so everybody is welcome to join
from whatever
area they're excited about the paper
from
just get in touch with us
so we can begin with a little
introductions and

warm-ups

um

blue how about you can go first and then

we'll pass it to the first author

so i'm blue knight an independent

research consultant from new mexico and

i'll pass to daniel friedman the first

author of this paper

thanks blue so i'm daniel i'm a

postdoc researcher in california and

yeah usually we have everybody except

for the first author introduce

themselves and then the first author

uh steps in and gives a little

background on the paper so it's fun it's

the first time that we've discussed one

of my papers on the stream

and uh we'll be talking about a bunch of

different aspects of it and it's awesome

to have you here blue with a lot of

expertise in collective behavior and a

bunch of other areas

the big question that motivated not just

this live stream but this whole line of

research in this paper

is

assuming that each agent is a quote from

the paper assuming that each agent

nestmate has only limited access to

incoming information

how can a group of active inference

agents solve complex group foraging

problems

so in the case of ants that local

information is going to be the pheromone

density the chemicals that it can

perceive its local umwelt but in the

case of information foraging it's what

we see or it's our perspective or the people who we can communicate to and so this is a topic that's come up in many different guises which is again when you have a multi-agent system using active inference to model each agents how do we think about the way that the perceptions and cognition and action of each individual active inference agents how do those compose or interact so that the group can have adaptive or interesting behavior the big question of the paper and also again just a big open question hardly one that gets answered in its finished form by the paper or by this discussion anything you want to add only that um i was really excited to i actually i picked this paper and i've been kind of pressuring daniel to do it for a little while maybe three months or so since i first read it so um i picked this paper because complex behavior complex group foraging problems it's really linked to my interest in collective intelligence and in scaling intelligence which is also why i'm drawn to active inference so i'm just curious to see your insight into ants and the deeper like you know when we see a paper we see only the surface the final product and so it's really going to be fun to take a deep dive into the little nuanced things that kind of maybe led you to the the birth of this paper great

visual with sort of what's on the surface and what's beneath the surface and in fact foraging is such a fun task to study not to get ahead of ourselves but it's one of the few tasks of the ant colony that you can see outside you just stand next to the nest entrance just like this picture which is taken while i was doing field work in arizona so there where all the ants are coming in and out of this is the nest entrance and so this species the red harvester ant *pogonomyrmex barbados* it's like there are ants coming in and out doing some trash work mid-work but mostly what you see on the outside is foraging and so what happens underneath well here's this painted ant with the three yellow dots so we can take a little tracer down underneath and we can see what is happening beneath the surface of the foraging the paper is active in france an active inference framework for ant colony behavior and we read the authors previously it was a really awesome collaboration so thanks a lot to to alec maxwell carl and axel who each brought really important insights here and two of the main aims and claims and then blue you can provide any thoughts in this paper we develop and implement a bayesian simulation model derived from the active inference framework that

captures some of the underlying dynamics
and emergent phenomena of ant colony
foraging behavior
that's what we did and the second point
is
by integrating the bayesian ant colony
perspective
of hunt and others
with stigmatic regulation and the
scale-free formalism of active inference
fields in glazebrook 2020
we provide an integrative framework for
behavioral ecological and evolutionary
modeling in ants
such an integrated approach could
facilitate the transfer of modeling
insights from ant colonies to human
design systems
and i guess back
so cool aims and claims they weren't
very forthcoming with or or you guys
weren't very forthcoming with the aims
and claims of the paper but i found like
lots of them like nested like little
gems in the paper so this is just a
couple of them but i think that this
paper was very um you know
on the bleeding edge we say instead of
like the leading edge cutting into some
some novel areas and making some new
connections that we haven't seen before
or at least i haven't seen before
cool there's what we did which is
write a simulation based upon
theoretical grounding and then there's
the fields that were integrated which
was a sort of bayesian forager
multi-agent perspective

stigma g which we'll talk about soon
and of course active inference
and that is going to
hopefully develop field of ant research
and

other areas that stand to learn from
either the bayesian multi-agent
perspective stigmargy or active
inference so that is quite a large swath
do you want to read the first part of
the abstract

sure

in this paper we introduce an active
inference model of ant colony foraging
behavior and implement the model in a
series of in-silico experiments active
inference is a multi-scale approach to
behavioral modeling that is being plot
applied across settings in theoretical
biology and ethology the ant colony is a
classic case system in the function of
distributed systems in terms of
stigmatic decision making and
information sharing

the second half of the abstract is
here we specify and simulate a markov
decision process model for ant colony
foraging we investigate a well-known
paradigm from laboratory and colony
behavioral experiments the alternating
t-maze paradigm to illustrate the
ability of the model to recover basic
colony phenomena such as trail formation
after food location discovery
we conclude by outlining how the active
inference ant colony foraging behavioral
model can be extended and situated
within a nested multi-scale framework

and systems approaches to biology more generally

how from that abstract do we forage ahead that's why we have a roadmap first in the introduction section

we talk about bayesian multiscale modeling and stigmargy

and then introduce the need and basis of an active inference model of the ant colony

in the methods section we talk about the foraging task specifically which is the alternating teammates but that's really just a map it's kind of like

minecraft or simcity like you can specify any kind of map you want we used an alternating t-maze for a few reasons

and then we talk about the active inference model for ant colony foraging specifically

we present some results

and we didn't have any live ant recordings in this paper so the results are more like schematic

they suggest what kinds of things can be measured and be discussed

for multi-agent active inference systems specifically the colonies foraging performance and swarm coherence metrics

and then in the discussion we talk about three scales of analysis and implication for the work

which is at the nest mate the colony and the population scale

and we use those words intentionally because they're all

ants someone says oh there's an ants on my windowsill and they mean a little

six-legged nest mate but
if the colony is the organism then
wouldn't the ant be the colony and just
it's a piece of ant tissue on your
windowsill
and then we end with a concluding remark
so it's a relatively sparse
road map but maybe we don't need that
many signposts to get where we're going
any road map comments
no good let's forage ahead
exactly exactly
um
the key words were
ants
foraging just
why even continue ants foraging active
environment oh yes that's right
behavioral modeling collective behavior
evo eco diva or ecoevo devo different
people order it differently and
stigmergy so these are going to be
keywords that
open many doors
so let's go into them
we'll start with sort of the grand
gaia level keyword which is eco evo devo
that's just how i say it but we might
have done evo eco devo in the paper
eco evo devo stands for
ecology evolution and development
so this is one way to see it with these
three interacting domains so it's like a
tri-disciplinary or transdisciplinary
approach
that is going to just like the name
would have you believe just like action
and inference come together in active

inference

evodivo is going to integrate across
these three fields so what's happening
as organisms develop

in their ecosystem and how evolution
shapes that

but it's also

and this is drawn from the gilbert at
all paper and a good text to read on it
is the abohif at all 2014 paper

it's actually a historical lineage

so those who study the history of
evolutionary theory might be familiar
for example with the modern synthesis of
evolutionary biology

which happened in the first half of the
1900s and integrated for example
population genetics and quantitative
genetics with evolution well

here's a cool timeline from the gilbert
paper

talking about how from the 1800s with
darwin as well as

antecedents

moved through sequence of really
important developments after the first
half of the 1900s so these are
post-1950s post

the discovery or elucidation of the
structure of the dna molecule

things like symbiosis

the way that microbial interactions
shape development

um a lot of cool things and on the
bottom here growing awareness of
beneficial symbioses

so eco evo devo is referring to a
historical lineage that is becoming

increasingly recognized as an important
integrative take on
the systems that we're a part of what
would you have to say about ecoevo blue
just that it's constantly um
i mean we're becoming so much more aware
of
the the different facets that go into
eco evo devo and um how interconnected
and fundamentally important
they are
in a triad
um it's like for me uh the clear like i
remember being at the neuroscience
meeting and i don't remember the name of
the professor but she was a toxicologist
from johns hopkins and had been working
in her field for i don't know like 30
years and this is like when they're
doing the microbiome sequencing which
was maybe like 20
12 20 13 somewhere around there and so
they're doing like the massive
microbiome sequencing and she got up and
said um you know we're gonna have to
redo all of the toxicology work that uh
i've done my entire life which it just
was so um overwhelming and um
i i it just really blew my mind to hear
such a renowned professor stand up and
say you know we don't know what's toxic
and what's not because we've not
considered the whole and so in in
relation to toxicity and ecology and
evolution and development like we
consider developing like you know of
course we consider children eating lead
paint and so forth and i think we're you

know we've looked at evolution but but
really that this the ecology of our own
survival is something i think that
people are just starting to to become
more aware of and attuned to
and there's a famous quote in biology by
debzanski nothing in
life makes sense except in the light of
evolution it's like but doesn't nothing
makes sense except in the light of
development and ecology and physics and
so on so it's kind of like
right we should have integrative
frameworks
whatever their english titles are so
that we can understand all the light
that's being cast on the phenomena that
we care about
and i'm reminded of the holobiont too
that chris fields brought up right all
of that
yes
what are
some of the ways that we can think about
the phenomena we care about well one of
the big key words here is collective
behavior that's one of the key words of
the paper so here's a lewis thomas quote
it is in our collective behavior that we
are most mysterious
individuals are collective too
we were never individuals
and
here's a nice map from wikipedia
of
complex systems some of the key terms a
little fractal with complex systems and
complexity itself and then like a

palette with the art

we have the different key terms like
systems theory and evolution and then
here's collective behavior up here so
collective behavior is one of the
fundamental pillars of complexity
because collective behavior is just
studying

systems interacting

what do you think about that blue
uh i want to say it in the next slide
sounds good

what is the difference between
collective behavior and collective
intelligence

right and so this is something that um
you know we've talked about offline but
really like when you look at collective
behavior and collective intelligence and
this was just a map i also pulled from
wikipedia's definition of collective
intelligence and so here i mean this is
clearly you know framed in a human
situation or is it
do ants have open source software maybe
we can talk about that
um but but collective intelligence you
know it's these three branches framed in
this image it's coordination
and cooperation which both of those come
into
the collective behavior
like play when i think about
fish schooling or or like you know birds
flocking right this is like a behavior
right good that we behave and coordinate
to to behave collectively uh for for
each other's mutual benefit and for the

benefit of the flock or the school or
whatever

but the only thing that's left out in
this triad is cognition and so
like what is cognition is that input
output or is is cognition intelligence
or what's the difference between
cognition and intelligence so for me
there's a like a very blurry
uh boundary between what is collective
behavior and what is collective
intelligence um and so i mean we've also
talked offline and so i just want to
know you know my thoughts on this are
evolving

and i'm curious as to maybe people that
are listening to this or maybe anybody
that wants to come onto our next live
stream and talk about it like if they
have strong opinions on collective
behavior versus collective intelligence
but i'd like to hear if you have any you
know

overwhelming thoughts

it's um

definitely two different perspectives on
intelligence and behavior there's sort
of like

some who would say that intelligent
behavior it's defined by its outcome
and it doesn't matter what's happening
internally and then there's sort of the
possibility that intelligence could
exist in the absence of behavior
now frameworks like we discussed in live
stream 28 with mental action
with mental
and cognitive processes as themselves

being action

and being modeled as action in active
inference framework that sort of blurs
the distinction but definitely that's
gonna be a question like

is collective behavior the same thing as
collective intelligence is intelligence
simply behavior

and then something i always wondered
about with collective behavior oh well
you know collective behavior groups of
people

okay but aren't people groups of cells
so

if collective doesn't mean anything as
opposed to individual then we're just
talking about

phenomena or behavior so you know let's
drop the adjectives that don't really
matter so what are we signaling when
we're talking about collective behavior
as opposed to saying yeah i'm talking
about individual groups you know
individual crowds of humans for example

i definitely think that um

you know there's

definitely if

if you could have no

you could have intelligence without
behavior of course because babies can't
do anything but like lay there in bed
and like observe you know like they just
you know can suck their fingers or
something but but there's some
intelligence there that doesn't have
necessarily an associated action

and a tremendous amount of intelligence
because they're learning and you know

building their own model of the world um
and so what is collective and i was
listening to the um internalism
externalism live stream like number 5.2
for anybody who wants to go back in time
and revisit that with me i was listening
to that last night and and it's like you
know what is the preferred level of
analysis right right like from the noble
paper like where do we get this you know
preferential level diagnostic uh and you
know we are as individuals collectives
collect collections of human and other
types of cells so so what is this
collective behavior versus collective
intelligence i don't know it's it's
something that's always mushy to me
there's a kantian perspective on that
for another time
one way that we can talk about a
specific type of collective behavior is
by introducing this topic of stigma g so
sometimes when people talk about
collective behavior in the first slide
of their talk they'll say you know
flocks of birds and here's a group of
people walking through a landscape and
those are examples of coordination
amongst agents but stigma g is actually
pointing out a really specific and
important different type of collective
behavior so here's from the wikipedia
describing that stigma g was introduced
by a termite biologist pierre paul
grasse
in the 1950s so it's a recent
term but of course people have seen this
phenomena before

and he defined it as
stimulation of workers by the
performance they have achieved no not
performance uh based benefits for jobs
but more
deriving from the words mark stigma and
aragon or work like an error the unit of
work

it captures the notion that an agent's
actions leave signs in the environment
and signs that it and other agents can
sense which determine and incite their
subsequent actions so here we're not
just looking at sort of interactions
amongst subunits of the birds looking at
each other or the school of fish but
actually the modification of the
environment through the actions of the
agents and then the ways that the
environmental conditions modify how the
agents behave

so

here is like the agents in feedback with
the environment
and what grassley was looking at was how
termites how one termite would pick up a
little clod of dirt and then put it
somewhere probabilistically and then
that would change the likelihood that it
picks up or puts down certain other
pieces of dirt later and so you could
imagine like one termite building a nest
by itself
or many happening at the same time
maybe even moving pieces of dirt in the
same way so stigmargy is actually a
mechanism by which extended cognition
occurs

and by which cognitive processes can be
scaled not just by increasing the
bandwidth or the type of interactions
laterally amongst agents
but actually by externalizing
the work
to
marks on the environment
and in the paper it was written here we
present the first active inference type
model of stigma g and the first active
inference model focusing on insect
behavior
so think about that there's been a lot
of navigation work in active inference
but when the mouse is moving through the
team maze for example or when the eyes
are scanning over the piece of text
which are active inference models that
we've seen
they don't leave traces
even when there's multi-agent active
inference simulations the interactions
have been lateral and the environment
has been taken as fixed so that's why we
brought
ecoevo in
because now we're thinking about the way
that agents are in feedback with their
environments and each other and how
stigma juice as a topic reflects that
this is pretty cool you added this
second slide on stigma g so what does it
show
i did so this is another um
just a image that i pulled off of um
wikipedia and it's taken from this is
someone's phd thesis stigmatic

collaboration a theoretical framework
for mass collaboration
and again it shows here coordination
cooperation and collaboration as opposed
to cognition
in this collective
activity or collective behavior model
and one of the things that i just really
thought um
was cool to point out
was on the right side under
collaboration okay
so collaboration is distinguished by
co-created emergent shared
representation
which manifests as creativity i thought
that was super interesting because
creativity applies
implies novelty in some way to me right
it implies like either a new use or or
or you know developing something that's
entirely new making new connections
um as well as co-created emergent shared
representations result in divergent
production
and evaluation
so how does this co-create an emergent
shared representation result in
divergence i thought that this was just
it was an interesting figure and all
these different like components and end
results are are just an interesting way
to think about sigmoidy and like can we
go through our environment without a
trace like we've all seen like murder
mysteries or like i mean how hard is it
actually to not leave a trace in in the
environment behind you even if it's

hairs or or fingerprints or whatever it
might be

so so none of us is without this leaving
a mark on the environment and what how
does stigmergy contribute to human niche
construction which were incredibly adept
at and and how do we learn from that the
traces that are left by others

we certainly stigmatically generated
this

presentation because i also drew from
the 2006 work of mark elliott
and um

elliott wrote

many questions arise when confronted
with the streamlined efficacy and
apparent lack of organization and
motivation of these new global
enterprises like wikipedia and others
that were being mentioned in 2006.
not the least of which how does this
work

stigmatic collaboration provides a
hypothesis as to how the collaborative
process can jump from being untenable
with numbers above 25
towards becoming a new driver in global
society with numbers well over 25 000.

pre-blockchain

free bitcoin

pre whatever it is that we're going
through now but these four tenants from
2006 really say a lot which just like
blue's image reflected

collaboration depends on communication
and communication happens in networks so
that's kind of collective behavior 101
that's something that you'd see in the

flock of birds or the school of fish
collaboration is inherently composed of
two components without either of
which collaboration cannot take place
social negotiation and creative output
if there aren't lateral social
interactions there isn't uh an
interacting group and without the
creative output then the productivity is
low so maybe systems like that can be
designed but probably not that
interesting or important
this is for humans collaboration small
groups 2 to 25 rely upon social
negotiation to evolve and guide its
process in creative output so that's
like the startup model or the small team
model
however
elliott's proposal
is that collaboration in large groups 25
to
however many
is dependent upon stigma g
you need to have if you have 25 or more
people in a meeting and it's an hour
meeting it's like everybody can be
taking notes on a shared document
everybody can be editing the wiki
everybody can be typing in the chat
but if people are talking linearly each
person's going to only be able to speak
for two minutes on average so just think
about how that scales up if you had a
thousand people working on a project
each person is only going to be able to
speak for a few seconds per hour so what
are the mechanisms by which instead of

just lateral ephemeral interactions
amongst agents
how can we modify
a single source of truth
in no sense
so that productivity can actually occur
and it's important to note that it's not
that social negotiation doesn't take
place when we're doing stigmargy in big
groups so it's not like wikipedia
editors aren't aware of each other as
social entities
it's just that as elliot wrote
negotiation takes a back seat in terms
of the creative drafting process most if
not all stigmatic wiki collaborations
have discussions associated with the
content being developed but it's
possible to contribute to wikipedia.org
for instance without discussing what
you're contributing to or creating so
that's kind of a cool insight
anything else to add on stigma yeah just
the group size is super interesting
because we are um
like i'm writing a paper on central
place foraging right now actually like a
unrelated paper but um it's interesting
this large groups of humans because like
we're looking at hunter-gatherer
societies and the formation of camps and
like what did human society kind of look
like way back um and and really the
estimate uh that's you know pretty
agreed upon in the field is you know 500
people in a camp which it like you know
it's like 100 families of five is like
the upper limit and like you know mostly

like at least um maybe 250 people for a
camp to
maintain right for it to stay cohesive
for it to stay bound together and i
think it's interesting because what what
was the stigmargic
process back in like the hunter-gatherer
foraging days like what did that look
like it's just interesting to to wonder
about and think about if it's dependent
on sigmergy i mean like now i think
about we stack rocks right like if you
you're marking a trail like everybody's
been out hiking and you see these
rock stacks it's like what are those
like when you first start seeing them
and then you see that they're trail
markers and you know i don't have to
know like no one told me that rocks are
stacked there to mark the trail but it's
like this is a clearly human signature
nobody else stacks rocks up that's weird
right and so you see these stigmatic
like things that are just left as traces
in the environment that could be used to
guide like large large groups of humans
across time even
yes and even in the definition of stigma
tree we saw marks and signs so semiotics
and biosemiotics comes into play
let's get
to the ants and potatoes
the eusocial insects
you social is a cool word we're not
going to go too much into the difference
of uh what is eusocial versus other
social forms
but some of the

exemplar eu social insects are ants
all of the around 15 000 ant species in
the world are eusocial
so they all live in colonies
there's no solitary ants and they live
in colonies where there's
you could call it a reproductive
division of labor
also honeybees are eusocial with a queen
and worker distinction
and termites are as well um most bees
are not eusocial so honeybees are a bit
of an outlier in that respect and just
like there are eusocial wasps not all
wasps are usocial
and there's sort of six images here
because i want to show that there's some
related species like ants and bees and
wasps are hymenoptera that's an order of
insects but termites are in a totally
different order more closely related to
cockroaches
and you social insects are able to
accomplish tremendous feats of stigmary
and niche modification
here's a leafcutter ant colony being
excavated by tiny little humans big
colony here's
a bee hive with honey and the cells and
the amazing geometry that they can
construct or that they can work with
once constructed by humans
and also very large nests for termites
so you social insects are really cool
how do they do what they do
well
i wanted to pull from a few slides from
my phd work which was on ants so these

are just some fun slides about ants and
um

if you want to ever jump in blue then go
for it but this was from some early work
um that was

at in the middle school

age range talking to people about ants
so one of the first things about ants is
there's no leader

on the left side we see the um

without getting into sex and gender

politics there are female

reproductives called queens usually

again a whole another topic why and

should they be called queens there are
males

who are sometimes called drones they
don't play

as active a role in the day-to-day tasks
of the colony and then there are sterile
females which we call worker nest mates
and the image on the right shows the
queen laying eggs

which are being nursed and tended to by
the nurse task group of workers

so she can just the queen can just pump
out eggs and the nurses are taking care
of it

and they're also feeding the queen and
then you have foragers outside of the
nest

making decisions for themselves so

there's no bluetooth there's no

long-range communication there's no hive
mind need we just have independent

agents making decisions for themselves

locally and there's no leader ant

how do they do that how does it work

without a leader

the way it works is that individual ants
make probabilistic decisions about
behavior based upon
the rate and type of interactions they
receive

so here we can start with just this ant
on the top left and it's walking along
and it starts encountering a high rate
of trash and so it thinks or acts as if
it thinks

i should start picking up trash maybe
the next one i see i'm more likely to
pick up

encounters a piece of trash can identify
that by the scent
picks it up

and can continue walking until it
encounters another scent maybe the
midden pile where there's a bunch of
trash piled up

walks along for some more time
before it doesn't encounter any more
trash and then reverts to a more
generally poised state

so this can be modified with a drift
diffusion model or response threshold
model

there's a lot of mathematical models of
how different rate and type of
interactions

probabilistically can update behavioral
tendencies but this is just a general
way that we can look at how different
ant behaviors play out we can always ask
what is the individual interacting with
and how are they responding to that
interaction

well how do you go from just responding
to interactions to adaptive behavior
the nature and what the adaptive
behavior looks like is going to depend
on the ecology

again eco evo devo

so here we see two possible ecologies of
food on the left is a colony where food
is found sparsely but in very clumpy
piles this is like a picnic in the
argentine ants in this case it makes
sense that there would be a trail
that might get laid down that helps
direct nest mates to the location where
another nest mate previously found the
food

however there's also ants that are
looking for food sources that might be
mobile like looking for live prey or
maybe just scattered dead prey but like
just dead insects in the desert bringing
them back and in those cases you're not
any more likely to find food where
another nest may found it and so in that
situation you see a lot more independent
foraging activity and you don't see a
pheromone trail so just because we're
talking about pheromones there's a lot
of different kinds of pheromones there's
nest marking and queen pheromones and
larval pheromones it's not just trail
for pheromones and they're not just
making these kinds of nests
though there are ecologies that favor
these types of collective algorithms
and then
across ecologies
there's this lose it to find it

component which is the balance
you know in the sense between explore
and exploit or between pragmatic and
epistemic values and goals which we talk
about in active inference

because

if

every single ant were to follow the
pheromone trail
then they could exploit locally but they
would have a challenge discovering new
food resources whereas if none of the
ants followed the pheromone trail
then it wouldn't be informative and so
it'd be a totally different game as well
so there's this balance between
following

the traces that i've been made before
and going off and discovering new
resources so that applies to seed
foraging and also information foraging
which is kind of

a

parallel that we'll be looking at again
and again so those four
uh pithy little statements there's no
leader it's about the response to
interactions and the way it looks
depends on the inter on ecology the
interactions with the ecology and that
across tasks there's this lose it to
find it balance between explore exploit
those

capture a lot of the dynamics of ants
the life cycle of an individual queen is
that she is also born as an egg and
develops into usually a winged
fertile form that's called a gyne like a

princess

she will mate with males and found a colony but there's other lifestyles for example that don't require mating with males or

in which case the queen doesn't found a colony by herself but rather breaks off with some workers there's a lot of diversity in the ant life cycles but once the queen is installed in the colony she pretty much just is fed and lays eggs many of them dozens to thousands or millions depending on the species

the other sexual

type of ants is the male the male lives to breed they have a larval development where they're tended to by extinct nurses

and then males usually a little bit leaner and meaner

they try to have sex they sometimes succeed and then they die so they only live for um

an adult phase that's very very short so these are the two kinds of sexual ants the female and the male queen and the male

but what we see when we're looking at ants out there in the wild is usually the workers

so workers start their life by being born as eggs and they have a different developmental trajectory than the queen even though they're both diploid females in most cases

and they can live for months to years depending on the species

and there's a process called temporal
polyethyism by which a worker once it's
born
the first tasks that it performs are
inside of the nest
and it performs nursing work so it's
like as if right after being born the
first thing you did right there in the
um the neonatal unit was take care of
other babies that's kind of like how the
ants work when they're middle aged ants
they tend to be doing nest work like
architecture and trash cleanup
and then
in their uh elder worker phase they do
foraging which is the most dangerous
task as well as the most um
water loss in certain ecologies so this
is temporal polyethylene so it's not
that ants are born into being a nurse or
born into being a forager all the
workers progress through a stereotypical
sequence of tasks
in fact that progression through a
stereotypical sequence of tasks by the
workers as well as the fact that workers
and queens are both from the same genome
in many cases a diploid female genome
is what
has led many to propose you social
insects as emerging models for
behavioral epigenetics
so we know that they
display fascinating individual and
collective behaviors as well as niche
modification and stigmargy
ants also display behavioral plasticity
they can learn and they have memory they

can also integrate senses like vision
and olfaction and vibration
and again this is a rising in the
context of similar or even identical
genetic backgrounds because there are
clonal ants so here's waddington's
epigenetic landscape
showing how this developmental ball
depending on how it bifurcates
epigenetically can lead to different
developmental outcomes like this first
bifurcation is are you gonna go the
fertile route with developed ovaries and
a different kind of thorax and wings or
will you go the sterile route at which
point about 85 percent of ants have what
are called monomorphic workers
whereas the minority of ants have what
are called polymorphic workers like
there's two discretely different or more
types of workers like majors and minors
so

a lot could be said on this of course
blue can i just ask a quick question
so

i see like flying ants around all the
time i'm not seeing like queens but are
there like species of ants that fly also
um there aren't flying workers there
aren't flying workers um but during
mating flights

you will see flying ants and those are
guys and drones flying around
looking to mate especially in the
southwest us where you are
got it okay so that makes sense that i
see flying ants i'm not crazy like they
are ants that are flying okay

just checking

um

people have used many different
approaches

to study

how ants work and how it is the case

that you end up getting all these

different behaviors out of the same

genome

one of the approaches that's been taken

is just not looking at any of the

molecular details but purely

investigating from a behavioral

ecological approach and

some of the critical work here has been

done by my phd advisor professor deborah

gordon

and so she has looked for many years at

the relationship between the statistical

regularities of the environment

and the adaptiveness of different

distributed algorithms so in

environments that have

as we talked about earlier like spread

out food resources you wouldn't expect

to see

pheromone trails forming whereas if the

food resources are distributed like this

you would expect to see that and another

example is if it's expensive to forage

then it's going to be in a default off

mode until it's stimulated whereas if

it's cheap to forage and maybe even

expensive to allow that territory to be

taken by different species you'd expect

to see like an on until disruption mode

and these search algorithms are not just

idle play things of miramicologists

they've been proposed to be a bridge
between organisms evolution and ecology
bridging the search algorithms that
exist on the internet on computers with
the kinds of search algorithms that
natural systems use because wouldn't it
be interesting if we could have adaptive
search algorithms

that
make all the trade-offs and succeed that
species do
and this is an awesome paper by michelle
lenon

um kind of like a meta-analysis
across

ants studying the characteristics of
their environmental niche and then also
their foraging strategy so this is a
great work to complement some of
professor gordon's work

behavioral ecology again is just looking
at behavior and ecology together so we
haven't talked about anything molecular
but

indeed people have studied molecular
components of this
from

2016 and well beyond

we can use evolutionary genomics
to investigate some of the genetic
underpinnings of ant behavior
evolutionary genomics gives us the
toolkits to compare genomes across
species so we can look at species that
have behavior a versus b
and ask if there are gene families that
expand or contract in those species
so that's about the composition of the

genome like what genes are present
or not conserved or unconserved gene
families and sequence changes in the
genome
there's also
transcriptomics so that's gene
expression analysis
where we can look at the same tissue or
different tissues
and how gene expression is differing
between the male and females
as well as between the reproductive and
the sterile females
we can use transcriptomics to
investigate the differences between
estimates so a common experimental
protocol is for example to collect the
brain or
the antenna from the nurses versus the
foragers and then ask about how gene
expression changes are associated with
differences in behavior
and tissue specificity is really
important you can't just blend up an ant
and then say something about how tissues
are expressing
genes
and then also
most recently there's been some of the
techniques developed in other species
like fruit flies
rna interference which is like a way to
knock down the expression of a certain
gene and the first
gene express gene
modifications and ants were published in
2017 with the mutants are here
and right now they've only been like

knock out which is easier to do than
knock in experiments but
the future of ant
genome editing will be quite interesting
so these are using genetic and
epigenetic and transcriptomic tools to
study and behavior
another approach that relies on neither
behavioral ecology or genetics although
of course it's related to them is
studying physiology
so in physiology we're interested in
similar types of questions how are
species similar and different how are
the reproductive and steriles uh of
females similar and different how are
the different sterile nest mates similar
and different
and
one of the most common approaches here
has been to study the concentration
of molecules in the brain like
neurotransmitters dopamine serotonin
which are common in us as well as well
as some neurotransmitters that are trace
amines and humans but really important
for invertebrate
neurophysiological function like
octopamine and tyramine
and there's been
a relatively epic history of drugging
ants in the lab
going back to at least the 70s
giving them opioids hallucinogens
stimulants thec
all kinds of drugs and more recently
some of the first works where we've been
able to provide ants with drugs in the

field

and this is a phylogenetic perspective

by cami at all

looking at which neurotransmitters have

been investigated across which ant

species so it's pretty sparse so we

don't have a total

picture but that's why we do

evolutionary biology so we can study how

the same molecules in species a b and c

play a role and then maybe we can have a

good hypothesis about how it might work

for species d

so clearly i have no background in in

ant you know species neurotransmission

but i do know quite a bit about

neurotransmitters and i'm curious i

haven't read the paper but um are the

the use of different neurotransmitters

species specific in ants or do they all

pretty much use the same

neurotransmitters to do the same things

they have the same

neurotransmitters but species might

utilize them differently

and that would be the whole cool thing

to study

let's

talk about the ant brain

so on the right side

is a pogo head and this is what the

dissected brain looks like it's zoomed

in

the brain at the same scale as the head

is very small

so ants in their head have a lot of

muscle for their jaw and a lot of

glandular tissue and so the brain is

taking up a relatively small fraction of
the head let's see this video will play
um okay i don't think the video is going
to play but

sasha

has made some very beautiful
immunohistochemical images
of three-dimensional amp brains
on the left side we have just one half
of an aunt's brain
that a picture that sasha took
and it's juxtaposed next to a 2014 paper
that

developed a new ontology for the brain
regions of the invertebrate brain
because invertebrate neuroscience was a
very discordant field in different
species people would have different
names for clearly homologous brain
regions

and so this paper in 2014 brought a lot
of clarity to the field because it gave
us standard nomenclature for talking
about different brain regions so
although the invertebrate brain uses a
lot of the same chemicals and families
of neuropeptides that are used in the
vertebrate brain it's not organized like
the vertebrate brain so there isn't like
a brain stem in a limbic system and a
prefrontal cortex there are different
brain regions so just to kind of name a
few

here's like the optic nerve coming into
the optic lobe

so the eye is like this sort of curved
area that sends signals into the optic
lobe and then similarly the antennal

nerve comes up from this side
and projects into these kind of grape
looking clusters into the olfactory lobe
so the layout is very different but
there are some functional and
computational analogies to vertebrate
brains
and then
this area up on the top is called the
mushroom body the middle is called the
central complex and the mushroom body is
known to play a very important role in
route memory
for example the work in cataglyphis which
is another type of desert ant
so different brain regions have
identified functions
and what's cool about the insect and the
ant brain is we can almost like
visualize the whole thing at once
because it's small
some neural changes have been found to
be associated with the
bio
uh
with the biobehavior of foragers so for
example
a lot of hormones that have been
associated with foraging activity and
solitary insects as well as hunger
activity in humans
have been found to be differentially
expressed in foragers so although
they're not foraging for themselves
they're foraging for the colony
it turns out there are some extremely
deeply conserved neural circuits and
peptides related to hunger neuropeptide

y

which is i think called neuropeptide ff
in humans

there's changes in gene expression which
again are found analogously in solitary
insects like fruit fly as well as
unsurprisingly in humans some of these
questions about regulating hunger and
about how the perception the
interoception of hunger
is linked to foraging activity like
get up and do something if you're hungry
these are deeply conserved beyond ants
even beyond invertebrates and in
invertebrates

and then

just like so many other behaviors have
been found to be the case changes in the
biogenic amine neurotransmitters in
their signaling properties and
metabolism have been established in
invertebrates

so these are like tyrosine and dopamine
over here serotonin octopamine and then
the enzymes that are kind of like the
freeways between the neurotransmitters
that transform them into each other
synthesizing and degrading
differences in the expression of these
enzymes which we determine with
transcriptomics

can

influence the levels so we can use some
technologies like hplc or gcms to
measure the amount of neurotransmitters
in the brain

and then we can do gene expression
analysis to look at the receptors and at

the metabolic enzymes as well

why does it all matter

because a forager doesn't know how much food the colony needs it doesn't have a dashboard telling it how many seeds there are or how many larva need to be fed or how much food is out there and so it is faced with this go no go decision stay home or forage which is influenced by environmental features that it may not have total awareness of that decision

is much like the decision that for example an investor might face with looking at even when they have total information or partial information you have to make some sort of decision so

foraging would be easy if you knew where the seeds were and you knew exactly when to go and you never made a bad decision and that's like sort of you know buy low sell high like if you knew what the price would do then

trading would be really simple but the whole question is about decision making under uncertainty not under certainty and so that

is the question to forage or not to forage

and that distinction will come up again and again

anything on that blue

yeah so since you didn't specifically go into foraging um i i know we've talked offline about um different strategies that ants use for foraging and um it's

the common interest that that we have
and so ants are central place foragers
for the most part with a nest but you've
given um you know where they they go out
central place foraging just in case
people don't know is that they leave the
nest they go get food and then they
bring it back and share the resources
with the people that are living in the
nest and you told me about ants that use
other kinds of strategies i like you've
told me about the feed as you go
ants that don't have a nest but but is
are they also central place foragers or
do ants ever use other strategies for
foraging other than than
that there are polydomis
ants so they have multiple nests and
they will so that's not center place
because they have multiple places
they're bringing it back to
there's ants like army ants that um i
guess it could be considered a central
place but it's a moving central place
um
nomadic
yes exactly so lots of kinds of foraging
behavior
finally we get to active inference which
is
an integrated framework for studying
perception and action of different
systems
this is what was written in the paper
active inference is a bayesian theory of
behavior that accounts for perception
learning decision making and the
selection of contextually appropriate

actions

all of which are cast as a form of

approximate bayesian or variational

inference in this framework perception

is modeled as the formation of posterior

state estimates

learning is cast as the process of

updating of priors

as well as other parameters

and decision making and action are

processes that compare the evidence for

various models about future states and

observations under possible courses of

action

so we're going to

continue this thread

of having a unified framework for

perception and action

and we're going to apply that to the ant

system

anything on active inference here

every every week what could we say

voyage ahead four days so so much to say

about active inference

um and the last

key word before we get into the paper

more specifically

is uh the behavioral modeling

word

so here was from the paper describing

how active inference is related to

behavioral modeling

active inference framework naturally

lends itself to modeling behavior of

different systems across scales

generally what changes across models

available in the literature are agent

action affordances so what what can be

done by the agent visual scanning versus physical movement

the semantics of the prior beliefs so what are the belief distributions over and the hyper parameterization the prior distribution over beliefs like are those fixed or are those learned

from the point of view of the dynamics i.e the message passing and update equations all active inference models are the same

and so we've seen all kinds of behavioral modeling frameworks and while many people have sort of agreed that there's like an input output structure there's some sort of events from the outside world that

go to the system of interest to the agent whether it's a computer program or animal and behavior outputs that's kind of where the agreement stops

there's many many different ways to carry out behavioral modeling and totally discordant simulation approaches so one of the aims of bringing active inference to bear here

is that we could actually have a framework that not just integrates the perception and action stages but puts in the ability to specify arbitrary generative models in between that cognitive sandwich and look at the patterns across different systems and different scales because they would be looking exactly the same

from a model outline perspective

so

just to tack onto that like one of the

beautiful things about active inference

is it contextualizes the interaction

between the agent and the environment so

there is no

behavioral modeling like that behavior

doesn't take place in a vacuum right and

and so one of the beautiful things about

active inference is that it's always um

you know considering like well how does

this plant grow and it's like well that

plant grows depending on how much sun

there is and how much water there is and

so there's always this perception action

loop when considering behavior which is

just um i think probably one of the

greatest things about the framework

here's

the um

paper

that we started on

that we ended up modifying to introduce

stigma g2

it's a paper the acquisition of

culturally patterned attention styles

under active inference

so at first glance it's like wait it's

it's literally a paper about cultural

styles of creating pottery designs

so here's like what the pottery designs

look like

there's

um it's kind of black and white art

on pottery

and then

the eye can see a local

set of cells
so like the focal point and the
immediate periphery
and motifs are visually detected
by looking just at
pottery
and so different kinds of patterns can
be inferred
by the visual scanning
over pottery patterns
and so
in the quest to resolve a culturally
informed uncertainty
we have these scanning paths
over motifs
so here it's like the motif was just
like this white
line it's like the eyes moved across and
it's like hey i'm kind of curious is
there is it going to be a motif that
looks like a tetris piece scan up nope
okay
and then the scanning happens like that
whereas other motifs have different
scanning paths so that's going to be
a lot like the foraging paths that we're
going to be looking at
but there's a few questions that we want
to consider like how is visual scanning
similar and different to ant foraging
in other words
how does epistemic foraging as well as
pragmatic foraging differ
when you have skin in the game or
exoskeleton in the game you have to move
your body visually you can scan and then
go back and then scan and go back and
you haven't left a mark on your

environment

so it's a foraging task

but it's not stigmatic because your vision doesn't modify the object of vision

and so there's the similarities and differences between physical foraging and visual foraging and then there's the question about how these visual and bodily foraging

um

phenomena

are similar or different than conceptual or informational foraging

so foraging is kind of this concept that links all these different pieces from

an animal physically moving on a landscape

to our sensory modalities like vision scanning a landscape to informational landscapes and there's similarities and differences across these

so we'll look at the some of the figures and formalisms the code and so on

figure one

has a layout of the map and the model

and so the map looks like a t

at the bottom of the t is the nest so

this is like the point where the ants

emanate out of when they start foraging

and then when they return

that's their end point of their foraging

trip

and in the classical teammates paradigm

food is placed on one of the arms of the

tea like the left or the right arm

in this case we used an alternating tea

base paradigm so the food would be on one end of the tea for some period of time and then it would switch to the other side of the tea so this is the the foraging task that we set up but this is just a map you could make it look like a labyrinth or it could look like anything in this simulation what happens is the ants start at their nest and then they implement a model which we'll look more at soon but the ant's action affordances are movement it can move in any of the directions that are surrounding it so a future model could have ant inertia or it could only move forward to the um sides like a pawn or something like that in chess it could rotate but we didn't keep track of the ants orientation we just said the cell is about as big as an ant and they can move to any of the adjacent cells and the ants able to perceive and act in those local that local environment and then we um use the following to parameterize the model so here we go a little bit blurry but there's a generative model which is just like in any other active inference model it's a generative model of outcomes and hidden states here the outcomes are the perceived pheromone and the hidden states are the

true concentrations that are being
inferred of the pheromone
the um the the factors that go into
the the markov decision process that's
happening inside of each ant
are um
it's
prior overhidden states
the a matrix which maps from the hidden
state to the outcome the b matrix which
is describing how hidden states change
their time
and then policy which is suggested here
to be influencing the transition between
different
states and the free energy minimization
which dictates policy selection
and then um we discretized the pheromone
into 10 categories so there was 10
possible concentrations of pheromone in
this model
so that's
kind of setup of the model is food is
going to be placed and alternating
between these two sides of the maze
ants are going to leave the nest and
diffuse around with no preference for a
direction
except when they find a seed they're
going to start depositing pheromone on
their return journey and then
they have a preference for walking
towards regions of increased pheromone
and it'll be shown that that is actually
enough to recapitulate some of the most
interesting behavior of ants like the
ability to form a trail to food and just
one interesting piece that

differentiates
the active inference model from
traditional reinforcement or
reinforcement driven
uh notions of foraging is in those
models the reward would be the seed at
the end uh the food itself
and that the cues
would be the pheromone trail along the
way the pheromone trail would be seen as
like a means to an end and the seed
would be like the goal
but in active inference the reward in
other words what the preference is over
is actually over pheromone density
and the seed is seen not as a reward but
rather a semiotic cue
to switch strategies
so active inference really inverts the
situation
by making the reward about the
moment-to-moment decision-making of the
ants
and then sees what is traditionally seen
as a reward the food itself as actually
just a cue to switch behavioral
strategies
so that's one of the most interesting
things that we figured out
so can i just comment in here really
quick
um just that that
this switching task teammates is like
it's a very simplified version of the
switching bandit problem so we discussed
the multi-arm switching bandit back i
guess it was maybe 23 i don't know if
you recall which live stream it was 25

maybe um but it was back i think before
the summer so um
maybe 25 or 23. so
that that live stream talks more about
like a stationary band problem and a
switching band problem but had
multi-arms like i think eight arms and
here there's a link there's only two but
i just wanted to draw the parallel
between the two kinds of simulations
it was um i just looked it up it was 24
with empirical evaluation of active
inference and multi-arm bandits this is
a two-armed bandit it's a two-armed team
maze but you can imagine
the different switching dynamics or
different number of arms so it
absolutely is related to that
let's go into a little bit more detail
on the formalism and the model itself
so here is describing the process of
variational inference
and specifically there's a q
variational distribution
that's being uh fit
as a function of the policy and it's
looking at the hidden variables
pheromone density
through from the beginning of time to a
given time step
and of policy so those are the kind of
pieces that inference is being done over
and then there's a generative model
of the outcomes the hidden states and
the policies so that's what the agent is
doing variational inference on
and that model doesn't need to be
jointly considered all by all it can be

factorized which makes it very tractable
to compute
and then given the factorization
there can be a free energy minimization
such that
actions are selected
based upon
the optimization of this variational
distribution
in a way that it's
selecting policies that are expected to
minimize the free energy
so that's in the appendix and these are
some of the formalisms and it is
minimizing the difference between the q
variational distribution hidden states
at time t
given policy
the difference between that variational
distribution and
this p distribution the actual empirical
distribution
which is over outcomes states also
condition on policy
so
there's again so much to say about the
active inference formalism but really
importantly
the calculations are conditioned on
policy so it's like we don't need to
calculate the exact momentum or
trajectory of the car
we need to pick a policy condition on
what affordances we have
and so that in a way
simplifies and sidesteps a lot of the
questions about wanting to describe the
world to the highest possible degree of

precision because actually what we want
precision on
is our policy selection so that's one
key difference is that conditioning on
policy
and then
this ability to do variational inference
on a simplified and factorized
representational distribution q
and then minimize the divergence the
expectation of the divergence
between the q distribution and the
empirical p distribution including
outcomes so those are some of the pieces
of active inference that help make it a
really scalable and tractable framework
here's the figure
in the um
appendix

so we talked about the markov decision
process and what some of these variables
mean

but here is like the center point right
with this is where the ant is located at
a given time

and then what it can perceive
and its action affordances as well
are the adjacent squares so just like in
the visual scanning model the i could
perceive the center point and the
adjacent cells here it's pretty much the
same code actually

that describes how even though there is
a computational

um

[Music]

object there's a variable that describes
the actual pheromone density across the

whole map just like there was an actual computational variable that described the whole pattern of the pottery we're able to use um a a remapping matrix that zeros out all of the other cells that aren't the adjacents so that is what ensures that each agent is actually doing a purely local decision making so we kind of take a per agent approach to masking such that its action and perception are constrained to only its local neighborhood so that could be changed you could have ants that are aware of ten cells around them but only act within one cell and those are the kinds of extensions to the model that could be done but we constrained perception of pheromone and action to just adjacencies so this is one thing that like i wish you guys would have played with um because what i would like to see is like okay is there something in my immediate proximity like i read like in the likelihood of remapping if there's nothing in the immediate proximity then the ant moves around in accordance with brownian motion as opposed to moving towards the pheromone and like i would like to see just an extra step of code if there's nothing in the nine let's look in the 25 and move toward the thing with the highest pheromone

like that's one like because okay it's
like a just a more distal like well like
a trace of ascent or something like that
yes it could be done
it could be done by modifying the code
so here's alex github
and just look alec chance and ants
probably a coincidence that it's in his
name the last few letters but you really
never know these days
and um the main pieces of code to look
at are ants dot py python
that's the main script ants dot pi
config.pi has a lot of the parameters so
if you want to explore
having just two ants in the nest or five
million ants in the nest
you can just change a parameter in the
config file and then when the ants stop
pi is run it's gonna like look at the
config file
to
run
and then figure one two and three are
just for reproducibility to exactly
reproduce the figures in the paper
let's talk about the pseudo code
so we can maybe go over the code during
one of the dot one of the dot two but
this is from the paper
to summarize the computational activity
of each forager at each time code
pseudocode is provided here
see the code for full details
the active inference model as
implemented here does not presuppose or
imply any specific neurocognitive
architecture for any specific ant

species

i believe that was uh some reviewers

comments

about what this did or didn't say about

what was happening inside of each ant

and again that's why it's so fun to talk

about active inference and these broader

philosophical questions while realizing

that actually we're just looking at how

sensory information comes in and what

actions come out and the internal

generative model can be a hypothesis

that's informed by cognitive

architecture but in fact it isn't that

cognitive architecture it's kind of like

black box modeling but then we can go

into the box and start building sub

boxes and hypotheses about what is

inside

but

we don't need to presuppose any specific

um cognitive architecture and in this

case the ant has no memory

and no anticipation so it's like kind of

like a one step

make a move towards the pheromone it's

not predicting or remembering what's

happened but what happens at each time

step is

it's like you know for each ant

first the ant perceives the local

environment that's the pheromone density

it then updates or optimizes its beliefs

about pheromone density

with respect to the variational free

energy

it then after updating its

distributional beliefs about pheromone

density

updates its belief

about action so relatively given the

affordances moving in different

directions

update my beliefs about how different

action policies will relate to expected

free energy

so it's like let's just say there was a

pheromone above but not as much below

that would be perceived in in step one

and updated in step two and then it'd be

like well moving up

will exp i can expect

to be more like in my preferred state

than if i were to move down

and then action is sampled

based upon the relative proportions

of um

the the relative free energies

of the different

action

affordances so that action is sampled

from that belief distribution action

selection

and then the agent performs movement

and

in the case of

it carrying a seed so having already

found the seed it deposits pheromone

otherwise it doesn't deposit pheromone

on the way out there are ants that

sometimes deposit pheromones both ways

but we were studying um the sort of

simplified case where the ant only puts

down pheromone on the way home

so can i comment really quick on the

neurocognitive architecture so again

like after listening to the internalism
externalism live stream 5.2 i believe it
was um
you know somebody asked that question
and maxwell gave this great answer which
i thought was really cool like okay
so where in the brain is the generative
model in this like internalism
externalism debate and that's something
that we see like with the visual cortex
mapping etc so where is the generative
model and maxwell's answer was really
cool like when you go to machine
learning and you don't like look inside
the computer for the generative model
like where's the print of the generative
model it's like software it's just
running so i i always think that that's
a really neat um
and sorry sorry maxwell if i like really
uh paraphrased you horribly because i
may have but but but there's no need for
the model to be
necessarily inside the brain mapping to
some specific brain region it can just
be software that's just running
yes figure two showed an example trace
of just a simulation
so the food starts off and um has not
been discovered yet so like the density
is low and it sort of starts here
pure no pheromone then it's it's kind of
follows like a zigzag and you can see
like the food was on the right side and
so there was a pheromone trail that was
forming
down
to the nest starting up on the top right

with the food
and then the um it switches to the left
side and so there starts to be pheromone
laid down
going towards the left arm
but what's reinforced is still the trunk
because that part is like good to have
high density on no matter what
and so that was our
simulation this was a colony with 70
nest mates over 2 000 time steps with
three switches of the food location
and
we didn't optimize the parameters
or try to like find the best performing
set or
um
it was really just to show that this
phenomena
of a pheromone trail leading to food
that can loosely track location of food
changing due to the natural decay
of the pheromone
so informationally this decay rate can
be thought of as environmental
perturbation or drift that counteracts
the reinforcing effects of stigmatic
convergence or stigmatic tightening
we were just like this is
the first pass model
that displays that phenomena
and it could be optimized to explore
how it works differently what if one arm
is longer or all these different pieces
but this is kind of what the model
outputs and then in the supplemental
videos there's a lot of videos there's
some some large file size gifs of

you can see the ants themselves
diffusing around and then once there
starts to be a trail they start to like
really start taking advantage of it
one of the
pieces that was most exciting to think
about was how do we measure foraging
performance

and this speaks to a common um
[Music]

area in studying collective behavior
which is like how do we think about the
relationship between individual level
behavior and group behavior again we
know that as we talked about earlier
like the individuals r groups and so on
but just looking at a two level model
where we have like agents and then the
collective

what are the difference between
individual and colony level traits so
colony level phenotypes also that's why
we use nest mate and colony level so
that we don't have to use individual or
group because nest mate is always just
going to be an estimate colony means
colony

level phenotypes are summary
measurements of emergent properties
in that phenotypes do not apply
to separate nest mates but they're
enacted by nest mates and these are good
citations to check out
for example each nest mate has bodily
phenotypes like the width of its head
every nest mate has a head with or has a
mid-gut gene expression profile
but colony phenotypes are those that

only exist at the colony level so there's only one measurement of those per colony like the total foraging performance or the average interest in internet estimate distance or the sex ratio of larva so which sex each larva is is nestmate level but this larval sex ratio only gets one measurement per colony and so we looked at two measures of colony foraging the number of foraging trips round trip completed through time and a swarm coherence metric so this is just the amount of foraging trips completed through time showing how as a function of the number of nest mates in the colony um you get different dynamics so for example 70 ants finds it faster the first return trip happens before the 500 time steps finds it faster and ends up getting more but potentially not you know multiple times is more so there's like some sort of a scaling relationship between the number of nest mates and the performance and then like look at 10. like it finds a few seeds this blue line but then it kind of flattens out so maybe the pheromone trail that the first few brought home just dissipates so then none of the other ones are really able to latch onto that trail so it has a very qualitatively different shape than the other ones

and then this is the swarm distance
coefficient which is like the average
distance between the nest mates
and so
just interesting that again it could
change maybe if you run it from work
time steps or
you have it in a different structure
than the team is but it's almost like
the distance increases as the ants
spread out
and then there's almost like a
stationarity
where the average distance stays
relatively similar so these are just a
few examples of the kind because you
have in the simulation at least you have
full information
on the location of every nest mate so if
you want to do an individual nest mate
level
phenotype
you can calculate that trivially but
then also there's this question about
how do you summarize collective foraging
performance
so any thoughts on that
so the choice to have full information
on the location of every nest mate is an
interesting one and um also like i
wonder if it's a parameter that you guys
tweaked or played with at all like what
if you just have knowledge of half of
the nesting locations
like you could almost simulate being a
behavioral ecologist the grad school
simulator
where it's like

the computer is simulating 100 nest mates but then you can only see 10 of them

or or each ant can only see 10 of them so we have full knowledge or the ants have full knowledge so that that was the question that i um have oh here we as simulators have the total trace of the simulation and one important piece that is not included in this was like interactions among nest mates they just passed right through each other in this simulation so that's definitely not too realistic and in nature the way that ants know whether they're heading inbound or outbound they can look at the polarization of the sun their step integration and the memory there's visual scene recognition and then there's also importantly interactions with other nest mates on the trail like if you're bumping into nest mates carrying food you're probably heading out and if you're holding food bumping into empty mandibled nest mates you're probably heading in and so they can combine all these kinds of sensory information to determine a lot more information than the kind of drift model that we instituted here well and and it just reminded me of this model that we just built with this foraging paper that i'm part of and um in that model all the foragers know the

location of all of the food a priori
right like so they already know where
all the food is so so but learning where
the food is is really like a whole
different model right like going to get
the food and modeling interactions like
you know network formation in foraging
patterns is different than learning
where the food is how we learn from
others this collective like learning and
that's something maybe that's the
difference between intelligence and
behavior like you can just
follow what other people do that's maybe
behavioral but but like learning from
others based on different cues and so
forth i mean that that's a that's a
maybe an intelligence metric
let's return to this

broader question about stigma g as a
type of niche modification and how it
plays a role in extended cognition
so this is some
nice writing by axel probably and
referring to some of his work
the representation of the environment
leveraged to solve the tma's problem
i.e the map
is not the one entertained by nest mate
brains
but rather the one
that is carved out of the environment
and leveraged implicitly by each ant as
they each forage alone together as a
colony leaving cues that nest mates can
use adaptively so i like that kind of
like collective behavior as being alone
together

like each nest mate is perceiving its own
umwelt its own local environment is all it perceives
yet the way that the colony approaches this problem is truly collective as well as truly stigmatic
in active inference terms the ants are endowed with only an extended generative model
an extended generative model is one where some of the model parameters are encoded
not by the agent engaging in inference itself but by the physical environment in which it lives
so that's kind of like a person maybe can have a few numbers in memory but now imagine if you're hearing a string of numbers and you're writing them down and then you're able to go back and interact or somebody else could have written the numbers down
you're going back and your generative model is functionally including things that are now part of the environment like they're outside of you so that really speaks to this integrating internalism and externalism
and it brings up big questions like what other kinds of systems whether they're currently modeled not in the collective behavior framework or whether they're being modeled as collective behavior but not as stigmatic collective behavior what other systems can be modeled with stigmatic active

inference algorithms

and then

what kind of individual level nestmate

level behavioral heuristics contribute

to adaptive collective behavior

in the context of what kind of

statistical regularities in the niche or

regimes of stigma g because just it's

not like stigma g equals adaptive you

could imagine the switching teammates

where they reinforce it too rapidly and

then it is inflexible

as the food changes location

so it's not like oh collective behavior

equals it's going to be adaptive or

stigma g equals adaptive

there's actually probably only certain

combinations of parameters certain

regions of parameter space

in which specific environmental

regularities can be capitalized upon

so

the idea that um

the ants are only endowed with the

extended generative model in the

environment and i know like that in the

paper they mentioned that memory is

something that

could be played with but i'm reminded of

the story of hansel and gretel so like

where hansel leaves a trail of

breadcrumbs behind them so that they can

find their way back home but um or a

trail of pebbles the first time that

were not eaten so you leave the trail of

pebbles it's like modification of the

environment but like if if you

can just remember the way

right like like that's that's an
interesting dynamic between
knowledge gained from environmental cues
versus knowledge that's inherent and
retained and and i know maybe you have
maybe more to say about
um ants and ant memory and where maybe
these memories or or
if there's some genetic memory for ants
in different species or in maybe even in
just single colony right like you've
talked to me before about colony
foraging behavior and even in the same
species the foraging behavior is
different and so
i'm curious as as to your thoughts on um
we don't have to go in right now maybe
for next week but your thoughts on on
ant memory and how that may come to play
an interesting question i think
species where the foragers head out
alone for longer distances
and need to find uh
pieces of prey
and bring them back
we'd expect them to have higher memory
whereas the extreme case of just the
pheromone follower
you know pheromone respector ant
maybe there's a very limited role for
memory and then the question empirically
would be
are there morphological differences in
their brain is the mushroom body smaller
are there certain genes that have
differential expression patterns
how do nurses and forager
differ in the expression of those genes

like maybe nurses don't need to have as much spatial memory

but foragers do need spatial memory or

step integration for example it's been

shown in multiple species that nurses

don't have a circadian rhythm they are

just sort of low-key working and

chilling 24 hours a day

whereas foragers have a circadian rhythm

that's strongly in tuned by the light

cycle

so

how

does memory sort of

start

and then how does

the temporal polyethyism and the

trajectory of what the nest mate needs

to do for the colony

how does that lead to the development of

nest mate level cognition

and then how have the regularities of

the niche selected for colonies that

succeed by virtue of being composed of

certain kinds of estimates that have

certain kinds of memory

this returns us to stigmergy so what was

cool about this

i just thought it was cool i mean i'm

always looking for you know intelligence

and behavior and how these two are

linked together but but this paper which

i found like

self-published on amazon but i think

it's part of a proceedings anyway it was

just like a neat kind of um like google

find uh it says stigmargy an

intelligence metric for emergent

distributed behaviors um and i'll just read this little piece of it it says the complexity of aggregated behavior often depends on sigmoidy stigma g occurs when behaviors by individuals modify the environment while being regulated by the environment state stigma has generally been studied for the forward problem predicting the consequences of local behaviors it is also applicable to the backward problem synthesizing local behaviors to fulfill a global need the concept provides provides an objective measure of intelligence for natural and synthetic systems a system's intelligence is measured by its amount of effective stigmargy it not only adapts to a changing environ environment but also modifies the environment to suit the system's needs and goals so i thought that this was just kind of a neat refreshing perspective on a lot of the things we've been talking about today

agreed um intelligence is all is often thought of as just being like inside of a brain or in one agent

when we know that that um doesn't take the brain body into account doesn't take the social into account and then even then that's where this paper came in was okay so then even social cognition let's say you integrate brain and body as well as a social

you're still just in this floating mob type simulation

how do we talk about the way that
interacting agents modify their
environment how that updates their
tendencies

that's where stigma g comes into play
and that's why it was an important thing
to do

well and we're increasingly seeing the
links drawn between cultural evolution
and collective intelligence because as a
species i mean we're still obviously not
discovering fire

and learning how to you know make
buildings right like we already know how
to do that that's why we're not like
living in these camps anymore and so
we've evolved and you know none of our
ancestors knew how to use an iphone but
maybe you know in nine generations none
of our none of our uh what are they
called successors that that's what it is
none of our successors will also know
how to use an iphone so it's just us
right now but everything that we do
we're leaving these stigmargic traces
behind in our environment that enables
us as a species to evolve or become more
intelligent for better or for worse
anyway big questions so there's some big
questions for next week that uh i'll
give you to think about

are you here

so what is the relationship between
collective behavior behavior and
collective intelligence and i'd love to
have other people's takes on this
what are other possible measures of
group synergy

or colony synergy how are colony-level
metrics retained in ants over time
in this model forgetfulness is critical
so do you think that the ability to
forget is as important as memory in
human or ant behavior
um is stigmar a measure of intelligence
or behavior or both maybe
and then does trust play a role in you
social insect synergy and i have some
more but i thought that that was a good
start
all great questions i would
yeah i think i got some dot one
questions i got some dot two questions
got some dot three questions but
um this is an open line of research the
paper was published only this year in
2021 so if this is something where you
think oh i have a theoretical
contribution or
i study archaeology and i want to learn
how stigmar might relate to those kinds
of models or oh i'm way better at coding
than this and i could make dot dot
improvement on the code
let's do it
so these will be fun things to discuss
blue thanks for all the
help on the xero in this fun
conversation we hope to see
many of you on the dot one and dot two
in the coming weeks on the 21st and on
the 28th
and tucker in the chat wrote
would love to hear and learn more about
memory and collective systems
totally agree

and uh when we think of like the von
neumann computer architecture it's like
where's memory it's like i mean there's
two spots where it can be right it can
be in the hard drive long term storage
or it can be in ram short term storage
and then that has even percolated into
how we think about short and long-term
memory in organisms
but now what about the niche
modification
what does it look like and how do we
think about it is it simply that like
data a is in short term and data b is
moved to the environment or
is something modified about the
environment and something modified about
the internal generative model such that
the information can be recapitulated
only through interaction with the
environment
how would we think about memory in that
case and that's also um
why ant colony optimization and swarm
computation
are part part of the broader umbrella of
unconventional computing architectures
because they do challenge
or move beyond
some of the sort of classical notions of
memory processing bandwidth
clock speed we didn't have to talk about
that
but those are things that
studying natural systems
makes us wonder about
fun times blue
see you soon

see you later see everyone next week bye